

ST PETE-CLEARWATER, FL, USA

WMO: 722116

Lat: 27.911N

Lon: 82.688W

Elev: 11

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12873

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	42.1	45.4	26.0	20.0	48.5	30.1	24.2	51.8	25.3	65.9	23.1	63.9	9.2	20	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.8	92.4	77.4	91.2	77.4	90.3	77.3	81.0	86.6	80.1	86.4	79.3	86.1	8.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.4	152.6	83.5	78.7	149.1	83.2	77.4	142.5	82.8	44.5	85.8	43.4	86.1	42.7	85.9	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.7	16.9	34.6	94.6	4.9	1.5	31.1	95.7	28.3	96.6	25.5	97.4	22.0	98.5		
(5)			30.5	82.4	4.1	1.9	27.5	83.8	25.1	84.9	22.8	85.9	19.8	87.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.1	61.3	64.6	68.2	73.3	78.7	82.1	83.0	83.4	82.1	77.2	69.6	65.0
	DBStd	9.32	7.89	7.30	6.30	4.61	3.20	2.36	2.23	2.17	2.55	4.95	6.01	7.67
	HDD50	17	10	3	0	0	0	0	0	0	0	0	0	4
	HDD65	433	165	89	44	5	0	0	0	0	0	3	28	99
	CDD50	8810	361	413	566	699	889	963	1021	1035	964	843	587	469
	CDD65	3749	50	79	144	254	425	513	556	570	514	380	165	99
	CDH74	33741	183	329	712	1620	3708	5405	6135	6392	5211	2950	734	362
	CDH80	11578	10	35	99	371	1255	2093	2401	2526	1844	810	104	30
Wind	WSAvg	8.3	8.9	9.0	9.7	9.4	8.5	7.5	6.7	6.8	7.9	8.7	8.5	8.5
Precipitation	PrecAvg	50.2	2.5	2.7	3.0	2.4	2.5	6.9	8.4	8.6	6.7	2.6	1.5	2.4
	PrecMax	69.6	5.5	11.6	9.1	8.6	8.9	14.1	15.2	13.6	13.6	6.6	4.8	15.6
	PrecMin	38.6	0.4	0.4	0.3	0.2	0.2	3.1	4.0	4.6	1.7	0.4	0.2	0.5
	PrecStd	8.0	1.5	2.3	2.4	1.7	2.1	3.0	2.6	2.8	3.1	1.4	1.1	3.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.4	83.6	84.5	88.2	91.4	93.7	93.4	93.5	92.9	90.6	85.4	82.5
		MCWB	70.6	70.5	70.7	71.5	73.9	77.0	77.5	78.4	77.1	76.6	74.0	72.0
	2%	DB	78.9	80.9	82.4	85.8	89.8	91.4	91.5	91.9	91.1	88.4	82.5	80.8
		MCWB	68.8	69.6	69.7	71.1	73.5	76.7	78.0	78.3	77.0	76.0	72.2	71.1
	5%	DB	75.7	78.6	81.0	84.0	88.1	90.3	90.4	90.8	90.0	86.5	81.0	78.6
		MCWB	67.2	68.2	68.9	70.7	73.3	76.8	78.0	78.3	77.1	75.2	70.9	70.2
	10%	DB	72.7	75.2	78.7	82.0	86.4	88.9	89.5	89.8	88.3	84.3	78.8	75.3
		MCWB	66.3	66.5	67.7	69.8	72.9	76.8	78.1	78.3	76.9	74.1	69.8	68.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.1	73.2	74.9	76.0	78.2	81.2	82.4	81.9	81.8	80.1	76.4	75.1
		MCDB	78.3	79.5	81.1	82.6	84.8	87.4	87.7	88.2	86.1	84.7	81.6	79.4
	2%	WB	70.9	71.5	73.2	74.6	77.0	79.7	80.8	80.7	80.3	78.3	75.0	73.2
		MCDB	75.4	77.5	78.8	81.5	83.9	87.0	86.5	87.2	85.4	84.0	79.9	77.7
	5%	WB	69.3	70.2	71.6	73.3	76.1	78.6	79.8	80.0	79.1	77.3	73.4	71.9
		MCDB	73.8	75.7	76.8	80.0	83.2	86.1	86.2	86.9	85.1	83.5	77.8	75.9
	10%	WB	67.3	68.8	70.0	72.2	75.3	78.0	78.9	79.2	78.3	76.3	71.8	70.2
		MCDB	72.0	73.8	75.4	78.8	82.4	85.6	85.7	86.3	84.6	82.2	76.4	74.4
Mean Daily Temperature Range	5% DB	MDBR	16.8	16.7	16.4	16.2	15.7	13.7	13.1	12.8	12.7	13.6	15.0	15.3
		MCDBR	17.9	17.8	17.5	17.1	16.9	15.1	14.1	14.0	14.0	13.7	15.2	15.6
		MCWBR	9.7	8.7	7.6	6.6	6.1	5.6	5.4	5.2	4.9	5.2	7.4	7.8
	5% WB	MCDBR	15.9	15.8	14.9	15.2	14.7	13.5	13.1	13.0	12.7	12.3	13.1	14.2
		MCWBR	10.3	9.3	8.8	7.8	6.2	5.7	6.0	5.6	5.3	5.7	7.8	8.8
Clear-Sky Solar Irradiance	taub		0.344	0.357	0.369	0.394	0.424	0.449	0.473	0.455	0.423	0.390	0.355	0.344
	taud		2.525	2.496	2.449	2.343	2.314	2.324	2.264	2.329	2.384	2.462	2.506	2.550
	Ebn at Noon		286	291	292	286	276	268	261	266	272	276	279	281
	Edh at Noon		29	32	35	40	42	41	43	41	37	33	29	27
All-Sky Solar Radiation	RadAvg		1121	1354	1719	2064	2193	1986	1878	1804	1638	1512	1228	1037
	RadStd		82	92	146	102	118	159	113	112	112	94	67	86

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.65	N/A	N/A	+0.65	N/A	N/A	N/A	N/A	+247	+184
(47) Regional (32 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-7	-83	+311	+216

Nomenclature: See separate page